

Short project — open clusters with Gaia DR3

One sitting, ~40 minutes with the Copilot agent. Everything comes from the archive.

Open clusters with Gaia DR3

data: Gaia DR3 cone searches for **Pleiades**, **Praesepe** and **M67**
(`astroquery.gaia`, table `gaia_source_lite`)

do: plot the proper motions, **cluster** the stars in (*pmra*, *pmdec*, *parallax*) to pick the members, then the HR diagram of each cluster and the age ordering from the turn-off

deliver: proper-motion diagram + HR diagram per cluster, member counts, distances, and the three clusters ordered by age

optional: isochrone fitting for a real age

How to work

- **Ask for a plan first** and read it: *“propose a plan — numbered steps, files touched, how each step is verified. Do not change any files yet.”* Strongest model for that turn.
- Then **one prompt per step**, and run the script after each one. A step that you cannot check is two steps.
- Paste real errors, say what must not change, and start a new conversation when you switch problems.

Checks along the way

- The Pleiades are at **136 pc**, Praesepe at **185 pc**, M67 at **~850 pc**. If a cluster comes out at 2 kpc, the clustering found the Galactic field behind it, not the cluster.
- Expect a few hundred to a thousand members per cluster, in one tight clump in proper motion.
- The turn-off moves redder with age: Pleiades → Praesepe → M67. That ordering is the result.
- The full `gaiadr3.gaia_source` table times out at the moment (DR4 preparation); `gaia_source_lite` with a cone search returns tens of thousands of rows in seconds.